

Mack Thompson

U.S. Citizen | mackthompson16@gmail.com | mack-thompson.com | linkedin.com/in/mackthompson1/

EDUCATION

Auburn University

May 2027

B.S. in Computer Science & Mathematics; Minor in Business (BET); AI Certificate

GPA: 3.71/4.00

- Relevant Coursework: Artificial Intelligence, Data Structures & Algorithms, Software Modeling, Discrete Mathematics, Digital Logic Circuits, Assembly Programming
- Involvement: Competitive Programming Team, Association for Computing Machinery (ACM), Δ XA fraternity

EXPERIENCE

OCV LLC

Oct 2025 – Present

Software Product Intern

Opelika, AL

- Partnering with engineers, designers, and product managers to research user needs and prototype full-stack applications for first responder wellness.

Auburn University – Samuel Ginn College of Engineering

Feb 2024 – Present

Undergraduate Research Assistant

Auburn, AL

- Fine-tuned open-source computer vision models for medical image classification, building and managing a 5K+ image dataset that improved diagnostic accuracy by 18%.
- Built and deployed therapy app with Flutter and Firebase, integrating GPT text generation for mental assessments.
- Working with Auburn's Advanced Vehicle Technology Competition (AVTC) team to implement machine learning optimization for energy consumption.

JupJewels.com

Jul 2025 – Sep 2025

Software Engineer (Freelance)

Opelika, AL

- Developed a custom Shopify site for a local company, increasing user conversion by 35% from the legacy site.
- Used Liquid and JSON to implement a DIY jewelry creation feature which became the best-selling product.
- Improved system performance by refining search indexing and streamlining email automation pipelines.

Kahua Inc.

May 2025 – Aug 2025

Software Engineer Intern

Alpharetta, GA

- Expanded an Azure managed automated testing framework by developing 17 new tests and 12 internal functions.
- Increased test development efficiency by 30% by engineering a debug tool that enabled hot reloading using the Roslyn Compiler, eliminating the need for driver resets during runtime recompilation.

BWH Sports Camp

Jun 2021 – Dec 2024

Founder

Atlanta, GA

- Founded and scaled a youth sports program to 55+ campers and 6 staff, generating over \$100K in profit.
- Developed a cloud-based automation system that monitored registration emails through Pub/Sub triggers, queried the OpenAI API with predefined context, and sent formatted replies automatically.

PROJECTS

World Order Forecast | *PyTorch, Node.js*

Oct 2025

- Integrated global data pipelines to construct composite country scores via Ray Dalio's eight metrics of empire strength.
- Developed a deep learning model with a 50 year recurrent window to model cyclical geopolitical dynamics.
- Achieved 60% accuracy using walk-forward leave-one-out validation on 30-year projection deltas.

Social Platform | *React.js, Node.js, SQL, WebSockets*

Aug 2024

- Built a full-stack messaging and calendar-sharing platform featuring a RESTful API, WebSocket live server, and responsive React front end for real-time communication.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Swift, C++, C#, SQL, HTML/CSS, Liquid

Frameworks & Libraries: React, Node.js, Next.js, Flutter, PyTorch, TensorFlow, Pandas, Selenium

Tools & Platforms: Google Cloud, Firebase, AWS, Shopify, WIX, Power BI, JSON

Concepts: Full-Stack Development, Cloud Computing, Machine Learning, Data Analysis, Automation, Version Control (Git), Software Testing, OOP, Responsive UI/UX Design, Backend, Frontend